



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026 - 27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No	: CHE F 312
Course Title	: Chemical Engineering Laboratory -1
Instructor-in-Charge	: Pradipta Chattopadhyay
Practical Instructors	: Pradipta Chattopadhyay, Jay Pandey
Hours	: M W – 3 4 and T Th – 7 8

1. Course Description:

The course comprises of experiments in three major areas of chemical engineering, i.e., fluid mechanics, heat transfer and mass transfer. The course involves experiments designed to cover the areas related to the theory of

1. **Fluid flow:** flow through packed and fluidized beds, major and minor losses associated with fluid flow, flow characteristics, continuity principle, boundary layer theory and fluid moving machinery,
2. **Heat transfer:** conduction, convection, radiation and heat exchange equipments such as heat exchangers (shell and tube, double pipe, plate type, finned tube) and condensers.
3. **Mass transfer:** Vapor liquid equilibrium (VLE), separation based on volatility, mass transfer equipments such as distillation column, packed bed adsorber, and mass transfer with reaction and other relevant concepts.

2. Scope and Objective of the Course:

The main objective of this course is to familiarize the students on different principles of chemical engineering through hands on experimentation. The experiments are designed to enable students to apply the theoretical background assimilated in the relevant chemical engineering courses to practice in a laboratory environment. Emphasis is laid on employing scientific methodology in practice: each experiment requires planning, data collection, calculations based on chemical engineering principles, critical analysis, discussion and presentation of the observed result. Additionally, the students will also learn the operation of laboratory scale scientific equipment.

3. Course Learning Outcomes

- Gain fundamental hands on experience working with Chemical Engineering laboratory conditions and equipments.
- Learn how to apply important Chemical Engineering theory for calculations and performing experiments.
- Learn how to get proper data, how experimental data can be analyzed and how suitable conclusions can be attained.

4. Text Book:

- o Lab Manual for Chemical Engineering Laboratory - I.

5. Reference Material:

- o Will be shared in Nalanda

6. Course Plan:

The following experiments will be conducted in the Chemical Engineering Laboratory during the regular lab sessions as per the time-table issued by AUGSD Office.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
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S. N.	Experiments	Batch
	Orientation & Introduction – Section – 1	5/8/2026 (Lab Hours)
	Orientation & Introduction – Section – 2	6/8/2026 (Lab Hours)
1.	Experiments on flow through fluidized and packed bed with water	Cycle-1
2.	Determination of losses due to pipe fittings	
3.	a) Determination of losses due to friction in pipes b) Experiment on flow through a helical coil	
4.	Verification of Bernoulli's theorem and experiment on discharge through Venturi, Orifice and Rotameter	
5.	Experiment on Reynolds Apparatus	
6.	Determination of Centrifugal pump characteristics	
7.	Determination on Reciprocating pump characteristics	
8.	a) Determination of the thermal conductivity of solids b) Experiment on the heat pipe	
9.	a) Experiment on double pipe heat exchangers b) Unsteady state heat transfer set up	
10.	Experiment on cross – circulation drying apparatus	
11.	Experiment on vapor in air diffusion	Cycle-2
12.	Experiment on vapor-liquid equilibrium setup	
13.	Experiment on differential distillation setup	
14.	Experiment on plate type and finned tube heat exchanger	
15.	Experiment on steam distillation setup	
16.	Experiments on shell and tube heat exchanger and Experiment on heat transfer in agitated vessel	
17.	Experiment on mass transfer with chemical reaction	
18.	Experiment on adsorption in packed bed	
19.	Experiment on humidification in wetted wall column	
20.	Experiment on drop wise and film wise condensation	



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
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7. Evaluation Scheme:

EC No.	Evaluation Component	Duration	Marks	Weightage	Date & Time	Nature of Component
1.	Exp. Report (Data Analysis & Visualization, Critical Thinking, Discussion)	-	(C1) $10 \times 6 = 60$ (C2) $10 \times 6 = 60$	40 %	Regular Laboratory Hours	Open Book
2.	Cycle 1(or 2) Lab Exam	120 min	45	15 %	29.09.2026 and 30.09.2026 Regular Lab Hours	Closed Book
3.	Cycle 2 (or 1) Lab Exam	120 min	45	15 %	23.11.2026 and 25.11.2026 Regular Lab Hours	Closed Book
4.	Comprehensive Quiz	60 min	45	15 %	20.11.2026	Closed Book
5.	Comprehensive Viva-Voce Exam	10 min	45	15 %	26.11.2026 and 27.11.2026	Closed Book
Grand Total			300	100 %		

C1 = Cycle - 1 (Exp. No. 1 to 10), C2 = Cycle - 2 (Exp. No. 11 to 20)

8. Grading Milestones:

- Mid-sem marks will be announced via Nalanda/email. This would be out of 105 marks (60 marks for report+ 45 marks for Lab Exam-Mid sem).
- Pre-compre total (PCT) will be announced via Nalanda/email. This would be out of 210 marks (**70 % weightage**) and will include all the components except comprehensive quiz and viva-voce exam marks.
- Final grade will be announced on ERP via AUGSD. No interim request in this regard will be entertained.

9. Description of Evaluative Components and Associated Policies:

9.1 Lab Report:

Guideline for Laboratory Report Submission

- At the beginning of the semester, all students will be encouraged to form a group of 3 or 4 amongst themselves. Please note that due to operational constraints, a group size smaller or larger than prescribed may not be feasible.
- On the day of conduction of the lab experiment, a datasheet with experimental results will be noted by each group member. Based on the observation sheets, the students will compile a report which should include the following:-
 - Aim & Objectives** of the experiment: 1 mark
 - Laboratory Observations:** 1 mark
 - Sample Calculations:** 2 marks
 - Results and Discussion:** 1 marks
 - Conclusions and Understanding:** 1 mark



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
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AUGS/ AGSR Division

- Students are encouraged to utilize the lab hours for carrying out sample calculations, data visualization and building insights for the discussion part of the lab report, particularly since both faculty and TAs are available live during this time to solve your doubts on spot.
- It's expected that every group will submit a lab report as a hardcopy (printout) at the lab and attach it to a master file maintained at the lab for this purpose. Please make sure you write name and ID of all group members on top of each page. You are free to use any platforms/software for report generation, data analysis and visualization. Use of technology is encouraged. ***Handwritten and scanned reports won't be accepted!***
- The students are encouraged to work along with their groupmates to draft the report. **However, the "Conclusions and Understanding" section of the report needs to be individually written by each group member.** This means that each report will have 3 or 4 separate conclusion and understanding (i.e., one by each student in the group)
- Lab report will be due one week after the completion of the experiment. Only those lab reports which are received on the due date will be evaluated.
- **Plagiarism:** As highlighted, you're requested to submit a professional quality digitally printed lab report. Plagiarism might be construed as academic dishonesty and your submitted lab reports may be subjected through plagiarism detection software(s). While we understand there might be a few technical phrases common in all the lab reports, hence there won't be any penalty for similarity up-to 20%. However, there would be 50% marks deduction for those whose reports are 20%-40% similar. If found more than 40% similarity, zero marks will be awarded for involved students. Decision of I/C will be final in all such cases.

9.2 Cycle-1/Cycle-2 Lab Exam:

- The mid/end sem lab exam will be a closed book lab exam, where the students will perform experiments at the lab on their own and submit a report in the designated time slot.
- The students will be asked to pick an experiment (through lucky draw) from a pool of experiments relevant to their cycle.
- A handwritten report (similar to the lab report) needs to be written and submitted for evaluation in given time slot.
- The experiments will be shortened to enable timely completion of the experiment and report writing.
- **No separate make-up for the Lab Exams will be allowed.**

9.3 Comprehensive Quiz:

- 9.3.1 A comprehensive quiz will be conducted on the designated date as a closed book exam. Questions and options (in case of multiple choice questions) would be randomized and it's in your best interest to focus on your questions in hand since the time would be limited.
- 9.3.2 The quiz will be based on the theory behind the experiments covered in the lab and may include question on concepts pertaining to the chemical engineering principles involved in the experiments.
- 9.3.3 **No separate make-up for the quiz will be allowed.**



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
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9.4 Comprehensive Viva-voce Examination:

- 9.4.1 Viva will be closed book and you will be tested on clarity of concepts related to the experiments conducted, background theory, experimental procedure, calculations, data analysis and applications.
- 9.4.2 A panel of all instructors will take the viva and give individual marks. Average of those marks would then be taken and communicated to you. Please note that we may not be able to give you individual break-up of marks in viva.
- 9.4.3 **Only genuine requests for viva make-up backed by paperwork/evidence/proof will be entertained. A prior email permission would be necessary. Decision of the I/C will be final in all such cases.**

10. Online Consultation Hour: will be announced during the orientation.

11. Notices: Notices will be provided in Nalanda.

12. Make-up Policy: As mentioned above, **make-up for selected evaluative components will be granted for genuine cases only with proper documentation**. For all such cases, prior permission of the I/C is compulsory. Decision of the I/C will be final in all the cases.

Pradipta Chattopadhyay

Instructor-in-charge
Course No. CHE F312